

QUESTION:

1. What are the differences between Native, Hybrid, and Cross-platform mobile application development, and when is each approach commonly used?
2. What is the Flutter Ecosystem, and how does it help developers create modern mobile applications?
3. What is the Software Development Life Cycle (SDLC), and why is it important in building successful applications?
4. How do Flutter development practices and the SDLC work together in creating real-world mobile applications?
5. Why should developers understand these concepts before starting an application project?

ANSWER:

- 1. Native:** Built specifically for one platform (Swift/Kotlin for iOS/Android). Fastest performance, full access to device features, but you write separate code for each platform. Used when you need top performance (games, camera-heavy apps, AR).

Hybrid: A web app (HTML/CSS/JS) wrapped in a native shell (e.g., Ionic, Cordova). Cheap and fast to build, but slower and less "native-feeling." Used for simple apps or MVPs with tight budgets.

Cross-platform: One codebase compiles to multiple platforms (Flutter, React Native). Near-native performance and look, big time/cost savings. Used by most startups and businesses today wanting one team to ship iOS + Android together.
- 2.** The Flutter Ecosystem is an open-source UI toolkit by Google powered by the Dart language, featuring a rich library of widgets and the package repository pub.dev. It helps developers create modern apps through a single codebase, instant Hot Reload, and native machine code compilation.
- 3. SDLC (Software Development Life Cycle)** SDLC is the structured process of building software, usually in stages: Planning - Design - Development - Testing - Deployment - Maintenance It's important because it ensures the app is planned properly, bugs are caught early, deadlines are realistic, and the final product actually meets user needs instead of just "coding and hoping."

4. Flutter + SDLC together Flutter fits naturally into each SDLC stage:

Design: Widgets help quickly prototype UI

Development: Hot reload speeds up coding/testing cycles

Testing: Built-in testing tools (unit, widget, integration tests)

Deployment: Easy build commands for both app stores So Flutter speeds up execution.

while SDLC provides the structure/process together they help teams build apps efficiently and reliably.

5. honestly, it's just common sense you wouldn't start building a house without knowing what tools and blueprint you're working with. Understanding these concepts upfront saves time, avoids costly mistakes, and gives developers the confidence to make the right calls instead of guessing as they go.